INCH POUND

MIL-PRF-49404A(CR)
19 MAY 1997
SUPERSEDING
MIL-I-49404(CR)
3 JUNE 1982

PERFORMANCE SPECIFICATION
BIAS VOLTAGE REGULATOR, CN-1559/UA

This performance specification is approved for use by USACECOM, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers one type of common module, Bias Voltage Regulator, CN-1559/UA, referred to herein as the "Regulator".

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454	- Standard General Requirements for Electronic Equipment
MIL-HDBK-781	- Reliability Testing for Engineering Development, Qualification and Production

(Unless otherwise indicated, copies of the above specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4d, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

USA CECOM

SM-D-771765	- Schematic Diagram, Preamp/Detector Bias Regulator
SM-C-771774	- Preamp/Detector Bias Regulator Outline Dimensions

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.3 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL

ANSI/ASQC Z 1.4	- Sampling Procedures and Tables for Inspection by Attributes
ANSI/IPC S-815A	- General Requirements for Soldering Electronic Interconnections

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(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, Wisconsin 53202).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.b), a sample(s) shall be subjected to first article inspection in accordance with paragraph 4.2.

3.2 Materials. The contractor shall select materials which are fully capable of meeting all the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The Regulator provides low-noise regulated voltage used for biasing a mercury-cadmium-telluride detector array as used in an infrared system.

3.4 Design. The regulator shall meet the form, fit, function and interface requirements specified herein.

3.4.1 Weight. The weight of the regulator shall not exceed 0.09 pound (1.44 oz) maximum.

3.4.2 Dimensions. The regulator shall meet the dimensions and dimensional tolerances of Drawing SM-C-771774 and connector pinouts shall be in accordance with Drawing SM-D-771765 to ensure that items produced IAW this specification accurately interface with existing hardware.

3.5 Performance characteristics.

3.5.1 Output voltage and output current. The regulator output voltage, loaded with an 8.25 ohm ±1 percent resistance shall be +3.000 Vdc, (+0.025, -0.050) at +23°C, ±5°C, and shall not vary more than +0.020, -0.045 Vdc over the temperature range of -54°C to +71°C. (see 4.5.1).

3.5.2 Ripple rejection. The regulator ripple rejection, loaded at the minimum and maximum input voltages (see 3.5.1), shall not be less than 50 dB over the frequency range of

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30 Hz to 20 KHz at the following frequencies: 30 Hz, 100 Hz, 1 KHz, 10 KHz, 20 KHz (see 4.5.2).

3.5.3. Output impedance. Output impedance shall be not greater than 0.1 ohm at a bandwidth of 30 Hz < f < 20 KHz when measured per 4.5.3.

3.5.4 Output noise. Output noise voltage shall be not greater than 27×10^{-9} vrms/$\sqrt{Hz}$ at 10 KHz, bandwidth ≤50 Hz, when tested per 4.5.4.

3.6 Environmental conditions.

3.6.1 Temperature shock. The Regulator shall not be damaged (see 6.3.1) by sudden changes in temperature between -54° C and +95° C. (For verification, see 4.6.1).

3.6.2 High temperature. The Regulator shall not be damaged by storage to +95° C or continuous operation to +71° C and shall operate for 30 minutes at +95° C. (For verification, see 4.6.2).

3.6.3 Low temperature. The Regulator shall not be damaged by operation to -54°C or storage to -62° C. (For verification, see 4.6.3).

3.6.4 Shock. The Regulator shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and the shocks resulting from bench handling inspections. (For verification, see 4.6.4).

3.6.5 Vibration. The Regulator shall not be damaged by vibration over the frequency spectrum at the specified g levels and amplitudes shown in Figure 1. (For verification, see 4.6.5).

3.7 Reliability. The Regulator shall have a mean time between failure (MTBF) of no less than 5,400 hours, θ_o. (For verification, see 4.7).

3.8 Preconditioning. Each Regulator shall be subjected to at least a 20-cycle preconditioning environment as specified in Figure 2 prior to Group A inspection. (For exception to this requirement, see 6.2.g).

3.9 Coating. Unless otherwise specified, the Regulator, its parts and components shall be first cleaned and coated with a corrosion resistant coating applied consistent with best commercial practices which will ensure proper adhesion and surface protection under the full range of environmental conditions described in this specification.

3.10 Soldering. Unless otherwise specified, electrical soldering shall conform to the requirements of ANSI/IPC S-815A.

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3.11 Nameplates and product marking. The Regulator, its parts, components, subassemblies and assemblies thereof shall be marked for identification and reference designation. For guidance, use MIL-HDBK-454.

3.12 Workmanship. All electronic parts and components of the regulator shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust and tool marks. Wires and integrated circuitry shall be protected from contract with or irregular surfaces and be shielded from shorting.

4. VERIFICATION

4.1 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.4)

4.2 First article testing. When specified in the contract, the first article inspection shall be performed by the contractor on a specified number of sample units. First article reliability (4.2.3) will not be required unless specified in the contract (see 6.2.b.3). Sampling procedures shall be in accordance with ANSI/ASQC Z1.4.

4.2.1 Visual inspection. All parts, components, dimensions, weights, markings and workmanship shall be visually inspected in accordance with Table I.

TABLE I. Visual inspection.

Inspection	Requirement paragraph	Test paragraph
Damaged components	3.4	4.2.1
Interfacing dimensions not as specified i.e., Height, maximum Solder/lead protrusion Pins, connector location Pins, guide location	3.4	4.2.1
Weight over specification	3.4.1	4.2.1
Coating not as specified	3.9*	4.2.1
Soldering not as specified	3.10	4.2.1
Marking not as specified	3.11	4.2.1
Workmanship not as specified	3.12	4.2.1

*In ASQC Z1.4, use inspection level S-3 or objective quality evidence.

4.2.2 Tests. Upon successful completion of the visual inspections specified in 4.2.1, perform the tests of Table II in any order.

TABLE II. First article inspection.

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Inspection	Requirement paragraph	Test paragraph
Output voltage and output current	3.5.1	4.5.1
Output voltage ripple	3.5.2	4.5.2
Output impedance	3.5.3	4.5.3
Output noise	3.5.4	4.5.4
Temperature shock	3.6.1	4.6.1
High temperature	3.6.2	4.6.2
Low temperature	3.6.3	4.6.3
Shock	3.6.4	4.6.4
Vibration	3.6.5	4.6.5
Reliability	3.7	4.7

4.2.2.1 Output voltage ripple. Provide an output load to the regulator per 3.5.1. The ripple rejection shall then be measured by modulating the input supply with a sinusoidal signal which provides a variable input ripple. Use the procedure of 4.5.2 to measure output voltage ripple.

4.2.3 First article reliability. The lower inspection MTBF (θ_1) shall be demonstrated in accordance with the following subparagraphs:

4.2.3.1 Inspection level.

- Temperature: +50°C, (+5 -0)°C
- Temperature cycle: None
- Vibration: 2.2 g's, ±10 percent peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be perpendicular to the plane of the printed circuit board.
- Equipment on-off cycling: Turn on and let the temperature stabilize. Hold the operating temperature for 3 hours, then turn off and let the temperature stabilize. This cycle shall continue throughout this requirement verification.

4.2.3.2 Parameters. The discrimination ratio θ_0/θ_1 shall be 3.

4.2.3.3 Failure criteria and test length. The inspections of Table III shall be performed at least once prior to and once after the inspection of 4.2.3.1. Additional inspection of Table III may be selected by the contractor. The results of all Table III inspections and any physical damage shall be used to determine the success or failure of the reliability requirement. In the

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event of a Table III inspection failure, the time interval between the preceding successful inspections and the failure shall not contribute to the total cumulative hours. Verification shall continue until the total unit hours together with the total count of relevant equipment failures permit either an accept or reject decision in accordance with Figure 5. Only equipment "on" time shall contribute to the total unit hours. No single equipment's "on" time shall be less than one-half the average operating time of all equipment "on" test.

4.2.4 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2.c).

4.3 Inspection conditions for verification. Unless otherwise specified herein, the Regulator shall be operated at +23°C, ±5°C. The input voltage shall be +9.5 to +10.0 Vdc. Unless otherwise specified, all inspection set-up and load resistor tolerances shall be ±5 percent.

4.4 Conformance inspection.

4.4.1 Group A inspection. Group A inspection shall be conducted at an ambient temperature of +23°C, ±5°C on all Regulators following the preconditioning of 3.8 and the inspection of 4.2.1. Group A tests listed in Table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Output voltage and output current	3.5.1	4.5.1
Output noise	3.5.4	4.5.4

4.4.2 Group B inspection. Not required.

4.4.3 Group C inspection. Group C inspections shall be conducted on Regulators selected from units which have passed the tests in 4.4.1. The sample(s) shall be tested in accordance with the inspections specified in Table IV. Samples shall be selected in accordance with the contract and ANSI/ASQC Z1.4 (see 6.2.d).

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Output voltage and output current	3.5.1	4.5.1
Output impedance	3.5.3	4.5.3
Vibration	3.6.5	4.6.5

4.4.3.1 Group C failures. Actions required relative to Group C failures shall be as specified in the contract (see 6.2.e).

4.4.3.2 Disposition of Group C samples. Group C samples shall be accepted on contract subsequent to the tests of Table III.

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4.5 Methods of inspection.

4.5.1 Output voltage and output current. Output voltage and current shall be measured to verify that the regulator is capable of supplying 60 detectors at 6 mA each or a total of 0.36 amperes. A resistive load of 8.25 ohms, ±1% shall simulate the detector load. The loaded output voltage shall be measured at nominal input (+9.6 and +4.8 Vdc), high input (+11.0 and +5.5 Vdc), and low input (+8.0 and +4.0 Vdc) voltage conditions. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Ripple rejection. With the regulator loaded (4.5.1), ripple rejection shall be measured by modulating the input supply with a sinusoidal signal which provides a variable input ripple. An oscilloscope shall be used to observe the ripple inserted on the input and the ripple appearing on the output. Ripple rejection is defined as:

Ripple Rejection = 20 log [Input voltage ripple/Output voltage ripple]

Ripple rejection shall be measured from 30 Hz to 20 KHz. (At the following frequencies: 30 Hz, 100 Hz, 1 kHz, 10 kHz, 20 kHz.) The input voltages shall be set at a minimum and maximum (3.5.1) for separate tests. Failure to meet requirements of 3.5.2 shall constitute failure of this test.

4.5.3. Output impedance. Output impedance shall be measured using a varying load network with a modulated input and an oscilloscope to measure the AC output current and voltage. Output impedance shall be measured at the following frequencies: 30 Hz, 100 Hz, 1 KHz, 10 KHz, and 20 KHz. Output impedance shall be defined as the ratio of the peak-to-peak output current change induced by the varying load network. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Output noise. Output noise voltage may be determined by measuring wideband rms noise voltage at the regulator output with a true-rms voltmeter using the test setup and test methods of 4.5.1. Since the noise output of the regulator is very low, it may fall under the minimum threshold level of a true-rms voltmeter. The measurement may then be performed using a low-noise amplifier, with a single RC roll-off, connected at the +3 Vdc output of the regulator.

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Using the voltage gain and signal bandwidth of the amplifier, noise shall be calculated using the equation as follows:

$$e_{noise} = \{[e_{out} \div (A \times (NBW)^{1/2})]^2 - [e_{amp}]\}^{5/2}$$

Where: e_{noise} = equivalent input noise seen at the regulator output

e_{out} = rms noise voltage measured at the amplifier output

A = amplifier voltage gain

NBW = noise bandwidth = $(\pi/2)$ x amplifier signal bandwidth

signal bandwidth low 3 dB point = 10 Hz maximum
signal bandwidth high 3 dB point = 50 kHz minimum

e_{amp} = equivalent amplifier input noise

Failure to meet the requirement of 3.5.4 shall constitute failure of this test.

4.6 Environmental tests. Environmental testing shall be performed in accordance with the requirements contained herein, except when a contractor's test plan showing equivalent methods has been approved by the Government contracting office. When the contractor's test plan deviates from the test procedures prescribed, the contractor shall be required to show where his own tests and methods for verifying the specified performance are equivalent. Also, measurements taken by the contractor during verification testing of all environmental requirements shall be accomplished using known testing apparatus such as an environmental test chamber, vibration test table, pendulum shock device or similar equipment for which the technical capabilities to properly simulate extreme climatic or environmental conditions are scientifically documented. In general, the methods for testing to verify compliance with all stated performance characteristics selected by the contractor shall be equivalent to recommended procedures specified herein.

4.6.1 Temperature shock. Verify compliance with the temperature shock requirement of 3.6.1. Use of the tests prescribed in 6.3 is recommended. Failure to meet the requirement shall constitute failure of the test.

4.6.2 High temperature. Verify compliance with the high temperature requirement of 3.6.2. Use of the tests prescribed in 6.3 is recommended. Failure to meet the requirement shall constitute failure of the test.

4.6.3 Low temperature. Verifying compliance with the requirement of 3.6.3. Use of the tests prescribed in 6.3 is recommended. Failure to meet the requirement shall constitute failure of the test.

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4.6.4 Shock. Verify compliance with the shock requirement of 3.6.4. Use of the tests prescribed in 6.3 is recommended. Failure to meet the requirement shall constitute failure of the test.

4.6.5 Vibration. Measure compliance with the vibration requirement of 3.6.5. Use of the tests prescribed in 6.3 is recommended. Failure to meet the requirement shall constitute failure of the test.

4.7 Reliability testing. Verify that the mean time between failure (MTBF) at least 5,400 hours has been attained. Use of MIL-HDBK-781 as guidance for test verification is recommended. Failure to meet the requirement of 3.7 shall constitute failure of the test.

4.8 Preconditioning. Verify attainment of the preconditioning requirement of 3.8 (see 6.2.g).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2.f). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory).

6.1 Intended use. The Regulator is intended for biasing an array in an infrared system (see 3.3).

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. First article tests are required.

(1) Time frame for submission of first article test reports are required.

(2) Time frame for approval of the first article test reports are required.

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(3) First article reliability testing, if required.

c. Disposition of first article samples.

d. Sampling plans for Group C (see 4.4.3).

e. Necessary actions by the contractor in the event of a Group C failure (see 4.4.3.1).

f. Packaging requirements (see 5.1).

g. The preconditioning requirement of 3.8 is required for procurement of all modules to this specification made directly for the Government. Implementation of 3.8 may be waived by systems contractors who order modules per this specification for use in the manufacture of complete systems or subassemblies thereof if these systems are subjected to equivalent preconditioning.

h. Environmental pollution prevention measures are contained in the packaging material specifications or provided by preparing activity for recommended disposability.

i. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

6.3 Definitions.

6.3.1 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of regular performance to values less than specified herein.

6.4 Environmental inspection procedures. The test methods and procedures of MIL-STD-810, listed below, are recommended, but are not mandatory for determining compliance with the environmental requirements of this specification.

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TABLE V. Environmental inspection procedures.

Inspection	Req. Para.	MIL-STD-810 Method	MIL-STD-810 Procedure	MIL-STD-810 Remarks
Temperature shock	3.6.1	503	I	Non-operating mode. Step 2 use -54°C. Step 1 and 4 use +85°C.
High temperature (see Figure 3)	3.6.2	501	I	Test simultaneously with paragraph 3.5.1, for final 30 minutes of step 3. For step 2 use +95°C and steps 4 & 5 use +71°C
Low temperature (see Figure 4)	3.6.3	502	I	Test simultaneously with paragraph 3.5.1, in place of Table III inspection. For step 2 use -62°C and step 4 use -54°C.
Shock	3.6.4	516.1	IV, V	Non-operating mode.
Vibration (see Figure 1)	3.6.5	514.1	I, VIII	Non-operating mode. Use Figure 1 (page 12) in place of MIL-STD-810 curve. Sinusoidal cycle time shall be 120 minutes per axis. Dwell time shall be 1/6 of the cycling time at each resonance.

6.5 Subject term (key word) listing.

Voltage regulation
Module
Infrared

6.6 Change from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army - CR

PREPARING ACTIVITY:
Army - CR

(Project 5855-0164)

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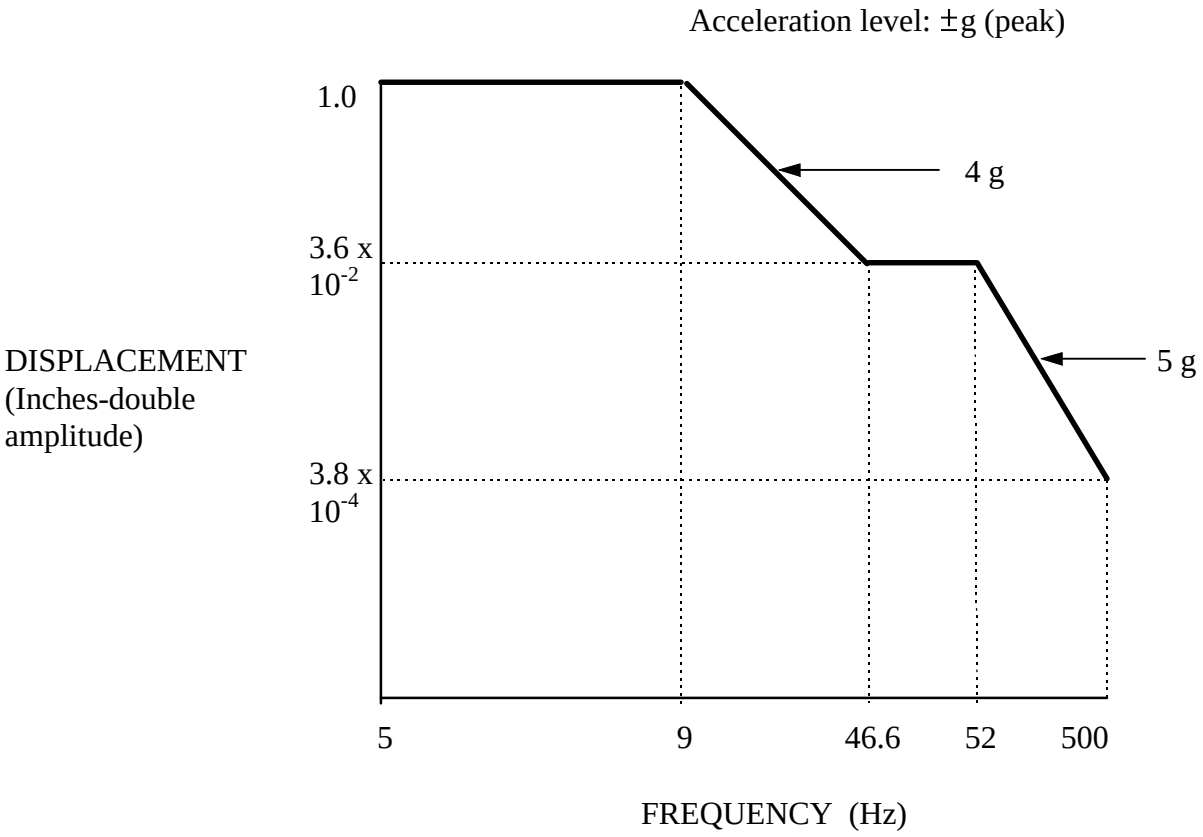


FIGURE 1. Vibration Test Profile.

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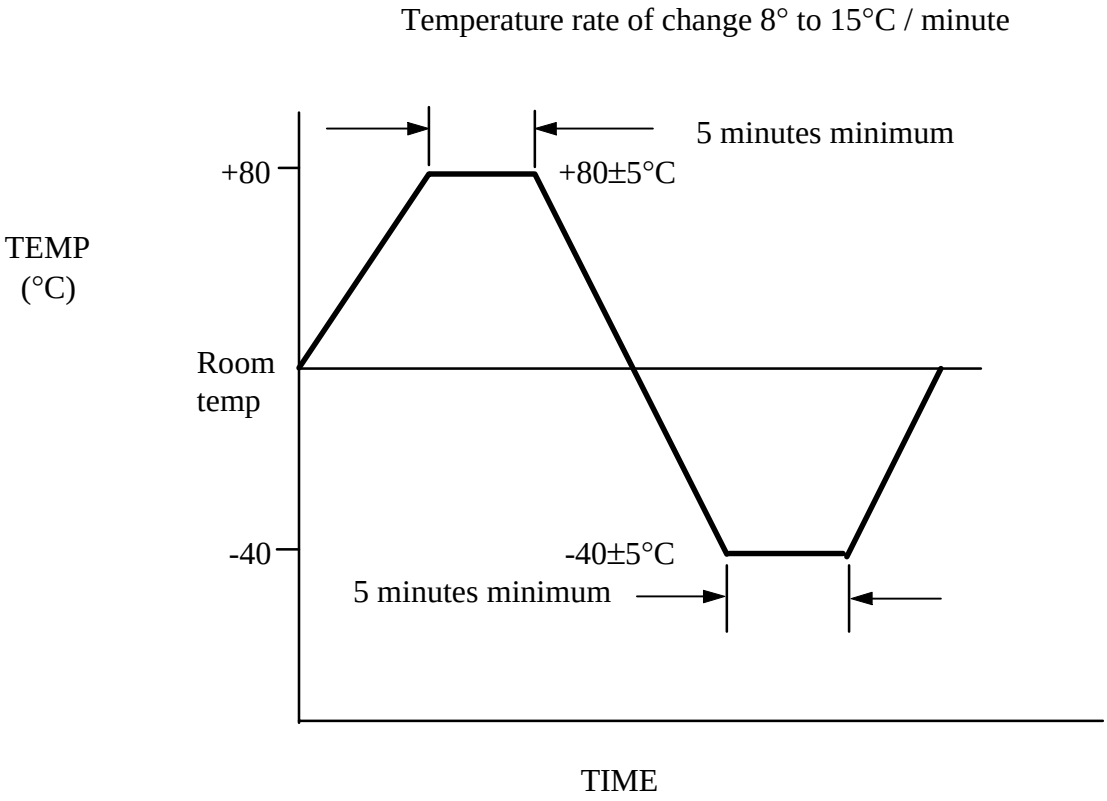
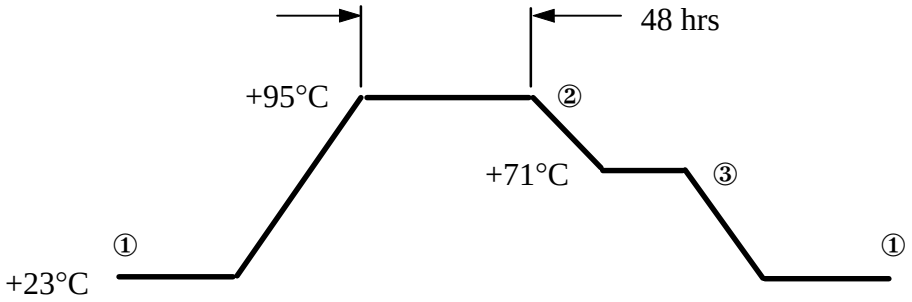


FIGURE 2. One Preconditioning Cycle.

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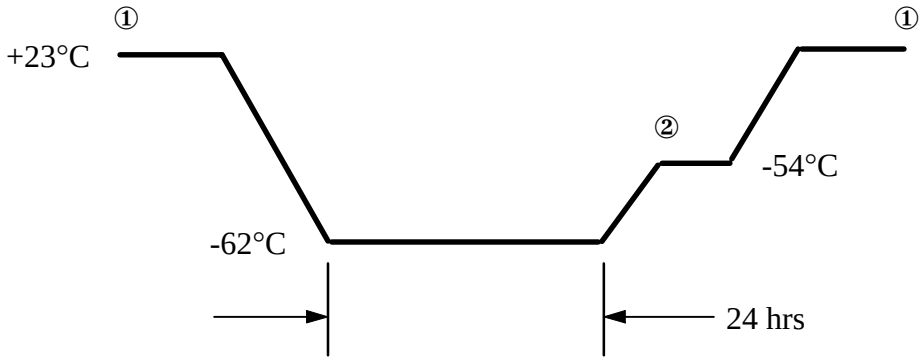
① Group A tests (Table III) are connected at these points.

② Test item is operated for 30 minutes at this point.

③ Output voltage and output current test (4.5.1) is conducted at this point.

FIGURE 3. High Temperature Test Profile.

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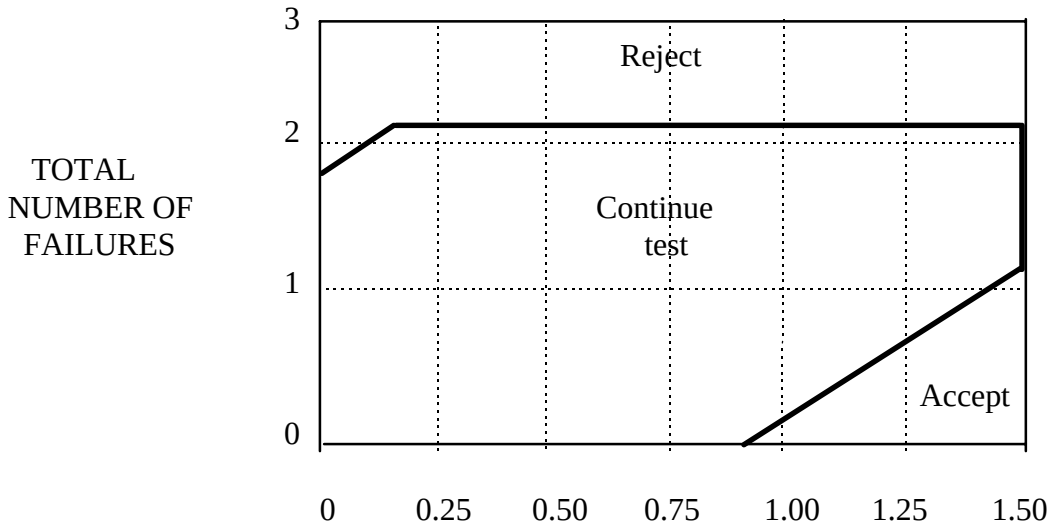


① Group A tests (Table III) are conducted at these points

② Output voltage and output current test (4.5.1) is conducted at this point.

FIGURE 4. Low Temperature Test Profile.

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TOTAL TEST TIME*

Number of failures	Reject (equal or less)	Accept (equal or more)
0	NA	0.89
1	NA	1.44
2	0.12	1.50
3	1.50	NA

*Total test time is the total unit hours of equipment “on time” and is expressed in multiples of lower test MTBF, θ_1

FIGURE 5. First Article Reliability Accept - Reject Criteria.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49404A	**2. DOCUMENT DATE (YYMMDD)** 970519

3. DOCUMENT TITLE BIAS VOLTAGE REGULATOR, CN-1559/UA

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ROGER LOUISAIRE	b. TELEPHONE *Include Area Code)* (1) Commercial: (732) 532-2735 (2) AUTOVON: 992-2735
c. ADDRESS *(Include Zip Code)* US ARMY COMMUNICATIONS-ELECTRONICS COMMAND, ATTN: AMSEL-LC-LEO-E-EP FORT MONMOUTH, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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